

Graphene wont replace Silicon

After demonstrating a graphene transistor running at 100GHz, IBM scientist claims that graphene will not replace the role of silicon in the digital computing regime

iPad to launch in India, finally

With the iPad2 about to launch anytime soon, Apple has decided to get their iPad to India, finally with prices around ₹33,000 for WiFi and upto ₹44,000 for 3G and WiFi combo

Enthralled @ IIT Techfest 2011

Nimish Sawant

nimish.sawant@thinkdigit.com

The year couldn't have started on a more technology heavy note. As if the Consumer Electronics Show (CES) 2011 in Las Vegas was not enough, Indian Institute of Technology Mumbai (IIT-B) had scheduled their annual technology extravaganza on the very same weekend. The IIT Techfest 2011, one of Asia's largest technology festival, was held in the IIT campus from January 7th to 9th. It was three days of technology overdose and we were left asking for more. The festival had many events, talks and paper presentations, various competitions involving robots and all the other usual activity happening at college fests. Unlike other college tech fests, the IIT Techfest is reknowned for its big ticket speakers and research innovations on display. This year saw lectures by the likes of Michael Jones - Founder, Google Earth; KS Pua - Inventor of the pen drive; Jaap Haartsen - Inventor of Bluetooth technology; Holger Bech Nielsen - the father of String Theory; Richard Stallman - from the GNU Project and Free Software Foundation among others.

We decided to concentrate more on the innovative tech on display at the campus. Presenting below, the best of the lot.

OpenViBE

French scientific organization INRIA had displayed the brain computer interface software

called OpenViBE. It is a software platform for designing, testing and using brain computer interfaces. What the interface essentially does is process the electrical signals linked with brain activity and convert them into a command or language that can be understood by the machine. The



OpenViBE system

brain activity is measured by a process called electro-encephalography (EEG). The user has to wear an EEG cap which has a number of electrodes located at strategic positions, which are linked to a device which helps the brain to communicate with the computer.

On asking about the practical applications, we were told that brain-computer interfaces along with OpenViBE could be used to assist patients with motor disabilities, in virtual reality gaming, home automation systems among others.

Willowgarage PR2 Robot

Just last month in Digit we had spoken on gesture based computing, about a Kinect hack which was being proposed to be used along with an open source Robotics Operating system

(ROS). Seeing Willow Garage's PR2 robot brought a smile on our face, as this was something familiar. Personal Robotics program is an endeavour to use robots in everyday usage scenarios. The Willow Garage community has come up with the open source ROS system and have given away around 10

to 12 PR2 robots to various universities, research organisations, etc, to build up on the ROS and come up with variety of applications for the robot. The idea is to have a common platform and share code information on the

innovative ways in which a robot can become as ubiquitous as a personal computer.

At the demo, we were shown the PR2 robot recognising various objects placed on a table and lifting it. The robot comes with a 3D camera built into it. Using lasers, the robot scanned the room with the right amount of depth and



Willow Garage PR2 Robot

displayed it on another monitor in real time.

The hardware that makes up the robot includes two quad core Xeon Core i7 processors, 24 GB of memory, two externally removable 1.5 TB drives and an internal 500 GB hard drive.

While still in the developmental stage, the PR2 robot can be utilized in areas like geriatric care, for doing repetitive tasks like doing the laundry, etc.

ECCE Robot

This was one of a kind robot we have seen. We have all seen quadri-pedal, even bi-pedal robots which mimic the human form factor perfectly. But the internal working of these robot is nowhere similar to the human body which involves bones, muscles joints, etc. ECCE Robot developed by the Artificial Intelligence lab of University of Zurich, University of Sussex, Technical University of Munich, comes closest to mimicking the way a human body works at the bone and muscular level. The robot on display at the Techfest was unfortunately not working

due to some issues, but we got a lowdown on its working. At the outset, the robot is a complex compilation of strings, elastic threads, motors and bone like plastic structures. Idea is to have a mechanism similar to that of muscles, tendons and bones while moving the robot.

Components making up this robot are easily available. For instance the bone is made of poly-